

RESEARCH ARTICLE

Machine Learning-Driven Crystal Systems Predictions of Perovskites From Elemental Properties

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ABSTRACT Predicting the crystal systems of perovskites accurately is essential for the design and accelerated discovery of functional materials. In this work, we have used machine learning (ML) algorithms to predict crystal systems of perovskites directly from the elemental properties of their constituent A and B site elements. A multiclass, multilabel dataset of 5,279 perovskite compositions was curated from the Materials Project and NOMAD databases, with feature engineering descriptors applied to enhance classification accuracy. Multiple balanced datasets were constructed to address class imbalances and polymorphism, and twelve supervised ML algorithms were evaluated using five performance indicators. The top performing models, like Extra Trees, LightGBM and Random Forest, achieved 88% to 96% overall classification accuracies across multiple datasets. Analysis of feature importance using SHAP values identified the top ten feature-engineered descriptors and intrinsic elemental properties that are primary determinants of crystal system prediction. This article highlights the transformative potential of ML-driven approaches in facilitating high-throughput screening, reducing dependence on costly experimental and computational methods, and deepening the understanding of structure-property relationships in perovskite materials.

INDEX TERMS Crystal system prediction, feature engineering, materials informatics, perovskites, SHAP analysis.

I. INTRODUCTION

The precise prediction of the crystal system of a perovskite by ML algorithms is revolutionising the discovery and design of novel materials. Perovskite materials are rigorously studied for their varied physical features, such as ferroelectricity, magnetism, superconductivity, and catalytic activity. Their multifunctional characteristics make them extremely attractive candidates for applications in photovoltaics, sensors, memory devices, energy storage, and spintronics [1], [2]. The varied behaviour of perovskite materials arises from their versatile crystal structures, capable of accommodating a wide range of cations with various sizes and valence states at the A and B sites, leading to multiple structural distortions and alterations in symmetry [3]. Optimal

perovskite oxides (ABO_3) comprise two distinct cations located at the A and B sites, with oxygen serving as the anion. The dodecahedral configuration of the A-site cation is surrounded by 12 oxygen ions, while the B cation forms an octahedral structure with 6 oxygen ions. They demonstrate ferroelectric, dielectric, and piezoelectric properties, mostly influenced by the crystal structure [4]. The crystal structure of perovskite-type compositions significantly affects the electrical configurations and impacts the energy band gaps and carrier mobility. The ideal ABO_3 crystallises in the cubic system. The deviation from the ideal cubic configuration is generally attributed to one of three factors. (a) cation displacement; (b) octahedral distortions; (c) octahedral tilting [5]. The first two pathways are affected by the electrical instability of the cation. The third mechanism results from the tilting of BO_6 octahedra while maintaining their corner-sharing connectivity. These structural anomalies

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produce tetragonal, orthorhombic, monoclinic, and triclinic crystal systems in perovskites [6]. Machine learning models provide high-throughput screening mechanism by rapidly analysing millions of theoretical compositions to identify those likely to crystallise in a specific system. This “in silico” pre-screening significantly diminishes the necessity for laborious and expensive experimental synthesis or complex *ab initio* calculations. In addition to prediction, machine learning elucidates concealed correlations between a material’s composition and its resulting crystal system, enhancing comprehension of the fundamental physical and chemical principles that dictate perovskite formation and structural stability. The machine learning model predicts the crystal system of perovskites based on their elemental properties, as ionic radii, electronegativities, oxidation states, and octahedral tilting influence the structural stability and crystal system prediction of perovskites [7], [8].

The computational techniques are challenging for compounds with extensive unit cells and functional cells significantly larger than primitive ones. Research in ML has created new avenues for accelerating materials discovery and design. Machine learning techniques can swiftly evaluate large datasets of elemental properties from established databases, expediting the assessment of millions of theoretical perovskite compositions that would be computationally impractical for conventional *ab initio* methods [9], [10]. Moreover, machine learning demonstrates proficiency in discerning intricate, non-linear correlations among various constituent parameters and the resultant crystal system, transcending basic empirical principles and uncovering the nuanced interactions that dictate perovskite formations [11]. This capability significantly expedites materials discovery by prioritising attractive candidates for diverse applications [12]. Furthermore, machine learning models can generalise to forecast crystal systems for completely new, unsynthesised perovskites and offer analyses of feature significance, emphasising which elemental characteristics are paramount in ascertaining the final crystal system, thus enhancing our fundamental comprehension of perovskite chemistry and crystallography [13], [14]. This innovative method renders ML-driven crystal system prediction from elemental parameters a cost-efficient and effective instrument, radically altering the design and discovery of these essential materials. The goal of this work is to demonstrate the efficacy of machine learning in accurately predicting crystal systems, elucidate the impact of chemical descriptors on the resultant structures, and ultimately furnish a predictive tool that can expedite the discovery of novel functional perovskite materials. The outcomes of this research could significantly decrease the time and expense linked to experimental trial-and-error and computationally intensive DFT simulations, hence facilitating the swift advancement of next-generation materials [15].

The following section of the paper emphasises the significance of material informatics, along with a concise survey of the literature on machine learning in the study of

various aspects of perovskites. Section III offers a overview of the dataset generation process, highlighting aspects such as data collection, feature selection, and dataset preparation. Section IV examines the various machine learning models and performance indicators used in this study. Section V conducts a detailed investigation of the performance of the ML models and the significance of features on various datasets. The concluding section presents final observations and examines potential directions for future research.

II. MACHINE LEARNING FOR PEROVSKITE PROPERTIES PREDICTIONS

Machine Learning evolved from the McCulloch-Pitts model of neural networks in 1943. It is widely used in many fields of research and industry to automate complicated tasks and make decisions based on data. These fields include robotics [16], computer vision [17], [18], fluid dynamics [19], [20], material science [21], and bioinformatics [22], [23]. Recently, ML has transformed into a cornerstone of artificial intelligence due to increased computational power, an abundance of big data, and state-of-the-art deep learning architectures. The “no free lunch” theorem in ML states that one particular algorithm will not perform well on problems in different domains, as their structures may differ. Convolutional Neural Networks (CNNs) [24] for computer vision, Recurrent Neural Networks (RNNs) [25], [26] for sequential data, and Transformers for modern large language models (LLMs) [27] like GPT, Llama, and Google’s Gemini are considered the primary deep learning architectures in ML. Other specialised neural network architectures that are commonly used for specific data types and tasks include Generative Adversarial Networks (GANs) [28], Physics-Informed Neural Networks (PINNs) [29], [30], Graph Neural Networks (GNNs) [31], Spiking Neural Networks (SNNs) [32] and Chaotic Neural Networks (cNNs) [33].

Recent developments in computational materials science have markedly transformed crystal system prediction, increasingly favouring data-driven methodologies, especially ML, facilitated by material databases such as the Materials Project (MP) and the Inorganic Crystal Structure Database (ICSD). Traditionally, *ab initio* approaches such as Density Functional Theory (DFT) combined with global optimisation algorithms (e.g., evolutionary algorithms) were employed for crystal structure prediction (CSP), implicitly determining crystal systems [34]. Although resilient, these are computationally demanding. The emergence of machine learning has expedited this trend. Feature engineering is essential for ML-based prediction, utilising descriptors derived from elemental properties (compositional features) to structural and symmetry-based information [9]. Diverse machine learning methods, such as random forests (RF), support vector machines (SVM), and neural networks (NN), are used for categorisation purposes. Random Forest, frequently using Magpie characteristics, has demonstrated robust efficacy in predicting crystal systems [11]. Convolutional neural

networks (CNNs) excel at analysing X-ray diffraction (XRD) patterns for the classification of crystal systems [35]. While GNNs have also shown promising performance in materials property prediction, they rely on detailed structural information, limiting their applicability in composition-only workflows such as the present study. Mao and Xiang's [36] recent surveys underscore the integration of ML into nearly every stage of perovskite materials research. Their analysis highlights that classical tree-based algorithms such as Random Forest, LightGBM, and SVM outperform deep learning when datasets are limited or heavily feature-engineered, such as in tasks involving crystal structure prediction.

A key trend is the integration of machine learning with high-throughput computational techniques. Machine learning models serve as efficient pre-screensers, significantly diminishing the search space for computationally intensive density functional theory calculations [12]. Machine Learning Potentials (MLPs), developed from *ab initio* data, expedite simulations by delivering rapid energy and force predictions, essential for investigating large configurational spaces and determining stable crystal structures [14]. Automated processes improve the efficiency of simulations involving Machine Learning Potentials (MLPs). Machine learning models can recognise complex, non-linear relationships that influence perovskite stability, predict stable and metastable structures, and efficiently evaluate compositional regions by leveraging elemental properties [37].

Several prior studies have applied ML specifically to perovskite crystal-system prediction. In a recent study, Behra et al. reported Shapley analysis and feature importance and classified perovskite oxides into crystal systems with an accuracy of 80.3% via a Light GBM model that incorporated structural, chemical, and geometric features [6]. Their dataset of approximately 800 oxide compositions relied heavily on structure-dependent descriptors such as bond angles and tolerance factors. Although their SHAP analysis offered useful knowledge about oxide chemistry, the absence of polymorphic materials and the dependence on structural information limit applicability to composition-only prediction. Furthermore, the relatively small dataset restricts the generalisability of their findings across broader perovskite chemistries.

In another study, Zhao et al. employed neural network and random forest models utilising one-hot encoded features, atom vectors, and Magpie to predict space groups and crystal systems with an accuracy of 85% [38]. Despite exploring diverse feature representations, their dataset contained fewer than 1,000 samples, limiting the robustness of deep-learning models. Their emphasis on predicting space groups rather than crystal systems, along with the absence of multilabel modelling, prevents the handling of real-world polymorphisms. A recent study by Xue et al. [39] proposed a deep-learning framework (TransMagNet) for predicting crystal systems and space groups of 102528 inorganic materials directly from chemical compositions with an accuracy ranging from 81% to 98% (overall accuracy

86%) depending on various crystal systems. However, the model treats crystal symmetry as a single-label classification problem and does not account for polymorphism. Metric learning techniques have significantly lowered computational demands by recognising structurally analogous templates for element substitution, attaining 96.4% accuracy [40]. Although computationally efficient, these approaches depend on structural prototypes and chemical similarity measures, making them unsuitable for early-stage screenings where only compositional information is available. They also contain highly distorted or metastable perovskites that deviate from known templates, limiting applicability in practical discovery workflows.

Template-Based Crystal Structure Prediction (TCSP) methodology was introduced by Lai Wei et al. [41]. It produces high-precision candidate structures ($\text{RMSD} < 0.1$) at scale by selecting prototype templates according to space group, oxidation state compatibility, and chemical similarity (EIMD). TCSP's online availability and its successful identification of hitherto uncharacterised stable compounds facilitate high-throughput discovery without necessitating substantial computational resources [41]. However, TCSP fundamentally relies on structural templates, making it unsuitable for large-scale compositional screening where millions of hypothetical A–B combinations may exist. Machine learning algorithms can facilitate focused synthesis, minimise trial-and-error experimentation, and explore metastable phases with innovative functionality by precisely predicting feasible crystal structures [14]. Machine learning is regarded as an essential tool for accelerating the discovery and design of perovskites, owing to its integration with burgeoning public materials databases such as the Materials Project and the Open Quantum Materials Database (OQMD) [42].

Composition-only machine-learning approaches, such as those employed in the present work, eliminate the need for structural inputs, significantly improving scalability for early-stage materials discovery. Compared to prior studies by Behra et al. (oxide-only dataset, 80.3% accuracy, no polymorphism handling), Zhao et al. (space-group prediction with small datasets), metric-learning approaches (template-dependent), and TCSP (structure-dependent), the present work introduces several key advances. It performs a systematic benchmarking of 12 machine-learning algorithms, employs a significantly larger dataset of 5,279 perovskite compositions across diverse chemistries, and incorporates explicit multilabel classification to capture real-world polymorphism. Additionally, the study develops 12 engineered features optimised for composition-only prediction and uses SHAP-based dimensionality reduction to derive a compact, physically interpreted feature set. Collectively, these developments establish this study as a scalable and interpretable ML framework for perovskite crystal-system prediction, effectively addressing gaps left by previous literature and enabling practical deployment in accelerated materials discovery workflows.

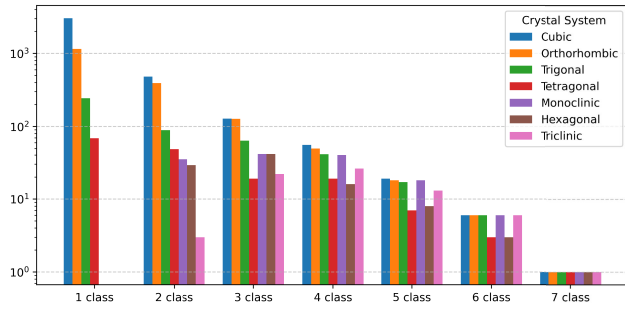


FIGURE 1. Grouped bar chart by class (logarithmic scale).

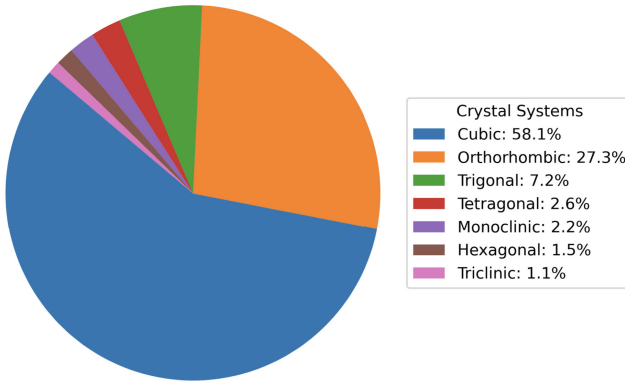


FIGURE 2. Distribution of crystal systems.

III. DATASET GENERATION

A. DATA COLLECTION

A multilabelled, multiclass dataset with formula, A-site element, B-site element, and crystal system of all available perovskites is collected from the Materials Project and NOMAD databases using their respective Application Programming Interfaces (APIs). 5279 perovskites obtained are classified into seven crystal systems: cubic, orthorhombic, trigonal, tetragonal, hexagonal, monoclinic, and triclinic. Due to polymorphism (one compound crystallises in multiple phases), the data is multilabelled, and so many materials form more than one crystal system. Among the materials in the dataset, 4502 were categorised into a single crystal system, 540 into two crystal systems, 148 into three crystal systems, 62 into four crystal systems, 20 into five crystal systems, 6 into six crystal systems, and one material (LaAlO_3) encompassed all seven crystal systems. Figure 1 presents a description of the multilabels related to materials and crystal systems.

Another challenge in this dataset is that it is highly imbalanced. Figure 2 shows the percentage of perovskite materials in each crystal system.

B. FEATURE SELECTION AND FEATURE ENGINEERING

The efficacy of ML models hinges critically on the selection and engineering of informative input features. For perovskite crystal system prediction, elemental properties serve as the foundational descriptors, translating chemical compositions into a numerical format amenable to algorithmic processing.

TABLE 1. Features added by feature engineering.

Features	Formula
Atomic radius ratio	$\text{ARR_A} = R_A/R_O$ $\text{ARR_B} = R_B/R_O$
Melting Point ratio	$\text{MPR_A} = \text{MP_A}/\text{MP_O}$ $\text{MPR_B} = \text{MP_B}/\text{MP_O}$
Electronegativity differences	$\text{END_A} = \text{EN_A} - \text{EN_O} $ $\text{END_B} = \text{EN_B} - \text{EN_O} $
Octahedral Factor	$\text{OF} = r_B / r_O$
Tolerance Factor	$t = (r_A12 + r_O) / \sqrt{(2)(r_B + r_O)}$
Bond length	$\text{BL_AB} = (\text{CR_A} + \text{CR_B}) - 0.09(\text{EN_A} - \text{EN_B})^2$ $\text{BL_AX} = (\text{CR_A} + \text{CR_O}) - 0.09(\text{EN_A} - \text{EN_O})^2$ $\text{BL_BX} = (\text{CR_B} + \text{CR_O}) - 0.09(\text{EN_B} - \text{EN_O})^2$
Electronegativity Ratio	$\text{ENR} = \text{END_B} / (r_B + r_O)$

Commonly utilised elemental properties are the ones that are related to structural stability, bond length, atomic and ionic interactions and advanced feature engineering descriptors. 25 features of A and B site elements are chosen in the initial phase. These include ionic radii with coordination 12 and 6 for A site and 6 for B site (r_{A12} , r_{A6} , r_B), electronegativity (EN_A , EN_B), Atomic radii (R_A , R_B), ionisation energy (IE_A , IE_B), density (D_A , D_B), thermal conductivity (TC_A , TC_B), valence electron count (VEC_A , VEC_B), melting point (MP_A , MP_B), boiling point (BP_A , BP_B), covalent radii (CR_A , CR_B), Van der Waals radii (VR_A , VR_B), and atomic mass (AM_A , AM_B). The literature provides several additional features, obtained through feature engineering, to enhance various properties of materials. 12 such features that are added to the dataset are given in Table 1. These engineered descriptors are physically motivated: atomic radius ratios reflect ionic packing and steric fit; melting-point ratios relate to relative bond strength and thermal stability; electronegativity differences capture ionicity and covalency trends; the octahedral and tolerance factors describe geometric stability; bond-length estimates incorporate electronegativity-corrected ionic radii following Pauling's principles; and the electronegativity ratio provides a measure of bonding asymmetry within the octahedral environment. Together, these features encode chemical and structural tendencies known to influence symmetry transitions in perovskite materials.

A correlation matrix of all the features is calculated. One of the highly correlated features ($\text{Icorr coeff} > 0.9$) among each pair is removed from the dataset while keeping the 12 features obtained by feature engineering. The resulting dataset has 27 features – ‘ARR_A’, ‘ARR_B’, ‘MPR_A’, ‘MPR_B’, ‘END_A’, ‘END_B’, ‘OF’, ‘t’, ‘BL_AB’, ‘BL_AX’, ‘BL_BX’, ‘ENR’, ‘r_A12’, ‘IE_A’, ‘D_A’, ‘TC_A’, ‘VEC_A’, ‘BP_A’, ‘VR_A’, ‘AM_A’, ‘IE_B’, ‘D_B’, ‘TC_B’, ‘VEC_B’, ‘BP_A’, ‘BP_B’, ‘VR_B’, ‘AM_B’.

A heatmap of the correlation coefficient between all the features is given in Figure 3.

C. PREPARATION OF DATA FILES FOR CLASSIFICATION

From the imbalanced multilabelled data file, we create the following data sets:

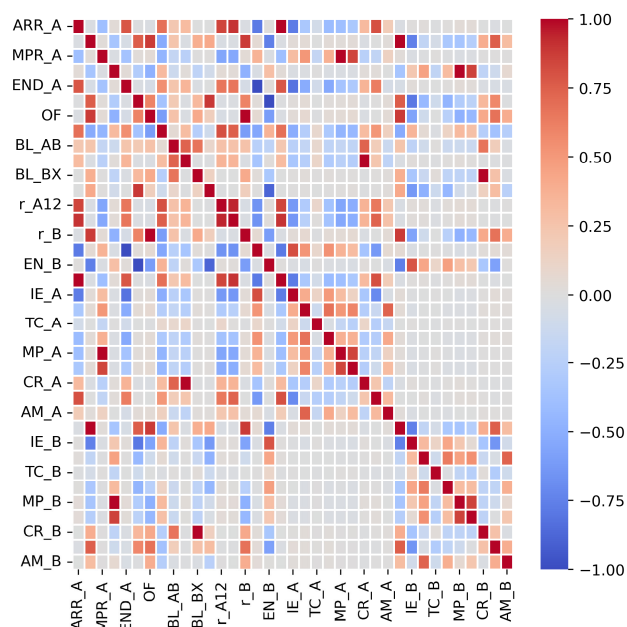


FIGURE 3. Correlation coefficient heatmap.

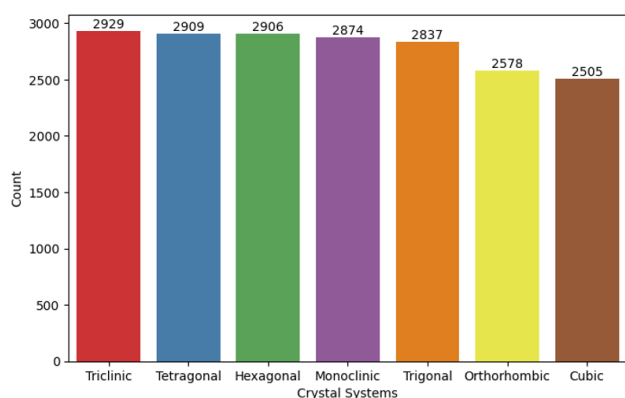


FIGURE 4. Samples per crystal system in M7 dataset.

- (i) A seven-class balanced training dataset (M7) consisting of 19,538 samples. This dataset was generated by applying an 80–20 stratified train–test split to the original data, followed by balancing the training set using SMOTE. Subsequently, samples with multiple labels (i.e., duplicate input feature vectors) were removed. The number of samples in each crystal system is given in Figure 4. Although SMOTE may generate interpolated feature vectors that are not strictly chemically realisable, its use here serves as a practical regularisation strategy to mitigate severe class imbalances and prevent model overfitting for majority classes. Importantly, all performance evaluations are carried out on a real, held-out test set containing no synthetic samples.
- (ii) A dataset M4 is made after removing multilabelled rows; the dataset is split into 80% training and 20% testing sets, and SMOTE is applied only to the training

dataset to address class imbalance. The test set contains exclusively real, non-synthetic samples. The balanced dataset (M4) has 9,712 training samples from four crystal systems—Cubic, Orthorhombic, Trigonal, and Tetragonal (2,428 samples each).

- (iii) To predict whether a sample (material) has a particular crystal system, the original data file was encoded with one-hot encoding for each of the crystal system classes. Here, the multi-label learning task is decomposed into multiple one-versus-all binary learning tasks, which are independent, one for each crystal system. To apply binary relevance, the entries with the same chemical formula are merged to allow for independent binary classification. 20% of the data from the encoded file, which contained positive and negative instances of all classes, was removed and kept aside to test for multi-class classification as unseen data. From the remaining 80% the binary encoded column corresponding to each crystal system was considered to form seven different datasets corresponding to each crystal system. All seven datasets were balanced using SMOTE. The number of samples in each of the balanced datasets is as follows: Cubic (5936), Orthorhombic (5600), Trigonal (7702), Tetragonal (8180), Hexagonal (8274), Triclinic (8322), and Monoclinic (8206).

IV. CLASSIFICATION ALGORITHMS AND PERFORMANCE INDICATORS

Given the labelled datasets of perovskite compositions and their corresponding crystal systems, various supervised learning algorithms can be used to learn the mapping function. Prominent and relatively simple examples include tree-based classifiers, sequential ensemble methods, Support Vector Machines, and Neural Networks. In this study, we evaluated twelve classification algorithms for predicting the crystal system across multiple dataset configurations: Logistic Regression, Naïve Bayes, K-Nearest Neighbours (KNN), Decision Tree, Random Forest, Extra Trees, AdaBoost, Gradient Boosting, XGBoost, LightGBM, Support Vector Machine (SVM), and Neural Network.

As the data is initially highly imbalanced, relying solely on Accuracy can be misleading. Metrics like Precision, Recall, F1-Score (macro or weighted average), ROC AUC, and the Confusion Matrix become critically important to ensure that the model performs well across all crystal systems, especially the rarer ones. To further strengthen the evaluation for the multi-label classification task, we added micro- and macro-F1 scores to capture both global and class-wise performance, the Jaccard score to quantify label-set similarity, Hamming loss to measure label-wise prediction errors, and subset accuracy to assess exact label-set matching. Stratified cross-validation is used to ensure each fold has a representative distribution of crystal systems. While not a direct performance metric in the sense of prediction quality, the time taken for model training and prediction is a practical

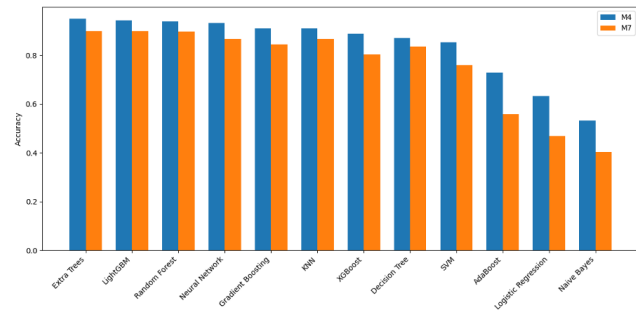


FIGURE 5. Classifier accuracies on M4 and M7 datasets.

performance indicator, especially when comparing algorithms or deploying models for high-throughput screening.

V. RESULTS AND DISCUSSIONS

The data in all nine data files were normalised using min-max scaling, and the data were split into 80% training and 20% testing data. The hyperparameters for the classifiers are tuned using GridSearchCV with five-fold cross-validation for each of the datasets separately. Each of the classification models was run ten times with different train and validation data splits.

A. ML MODEL COMPARISONS OVER VARIOUS DATASETS

Table 2 and Table 3 present the average performance of each of the classifiers on the M4 and M7 datasets.

The performance analysis of the classifiers on the seven-class and four-class datasets is given by the bar plot in Figure 5 that plots the accuracy of classifiers for each dataset. Tree-based methods, like Extra Trees, LightGBM, and Random Forest, were more robust and accurate on both M4 and M7 data files.

Since the evaluation was performed over ten independent runs, the mean and standard deviation ($\mu \pm \sigma$) of Accuracy, F1-score and ROC-AUC in M4 and M7 datasets were evaluated. The results are presented in Table 4. Pairwise Wilcoxon signed-rank tests were conducted at a 5% significance level to assess the statistical significance of performance differences between models. No statistically significant differences were shown in performance between Extra Trees, LightGBM, and Random Forest ($W = 0.0$, $p = 0.0625$). Thus, we conclude that all three ensemble models exhibit statistically equivalent predictive performance across cross-validation folds.

The consistent superiority of tree-based models across both datasets is supported by their higher F1-scores, precision, recall, and ROC-AUC values, indicating improved accuracy and balanced classification performance across classes. The strong performance of these models is due to their ability to effectively capture complex, non-linear, and threshold-driven relationships inherent in crystallographic descriptors and empirical stability rules. Ensemble tree-based models appear particularly effective at learning sharp, rule-based decision boundaries, which are characteristic of crystallographic stability fields governed by threshold-like constraints.

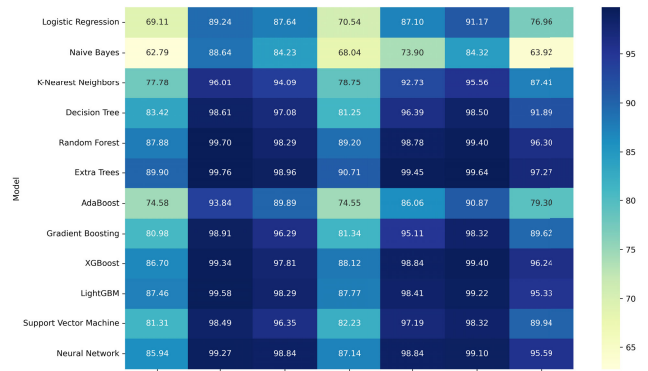


FIGURE 6. Model performance per crystal system (accuracy %).

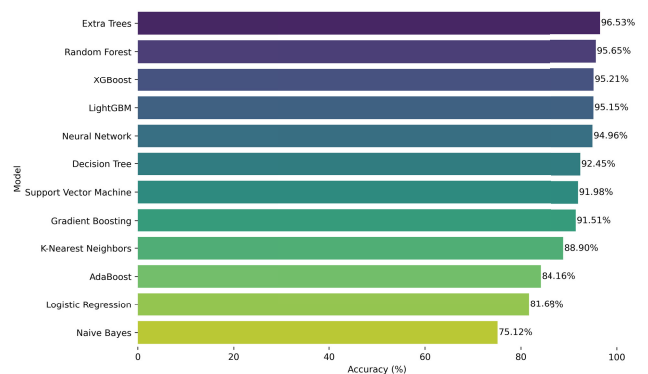


FIGURE 7. Average model performance across crystal systems.

Neural network models generally require substantially larger datasets to fully exploit their representational capacity and may therefore be over-parameterised for the dataset sizes considered in this study. Further, the weaker performance of linear classifiers indicates that the relationship between elemental descriptors and crystal system classification is intrinsically non-linear, favouring modelling approaches capable of capturing complex feature interactions.

Binary classification was done on all seven datasets formed from the 80% data of the encoded file. The test accuracies of the ML algorithms on all the datasets is shown in Figure 6. The average performance evaluation of the algorithms on the datasets corresponding to all the crystal systems is summarised in Figure 7.

Tree-based and boosting algorithms, along with Neural Networks, gave comparable results for the binary classification problems. A multilabel classification was done on the unseen 20% of data that was split from the original encoded file using the best three classifiers, namely Extra Trees, LightGBM and Random Forest. The classifiers performed extremely well on the dataset, and the results are tabulated in Table 5.

B. FEATURE IMPORTANCE AND ANALYSIS

SHAP (SHapley Additive exPlanations) values illustrate the prediction of an instance by computing the contribution

TABLE 2. Performance of classification algorithms on M4 and M7 datasets.

Model	Accuracy		F1 Score		Precision		Recall		ROC AUC	
	M4	M7	M4	M7	M4	M7	M4	M7	M4	M7
Extra Trees	0.9501	0.8771	0.9498	0.8768	0.9497	0.8765	0.9501	0.8748	0.9953	0.9779
LightGBM	0.9421	0.8832	0.9419	0.8830	0.9419	0.8827	0.9421	0.8807	0.9932	0.9859
Random Forest	0.9376	0.8731	0.9373	0.8728	0.9372	0.8725	0.9376	0.8703	0.9931	0.9827
Neural Network	0.9325	0.8569	0.9319	0.8562	0.9321	0.8556	0.9325	0.8531	0.9875	0.9784
Gradient Boosting	0.9108	0.8363	0.9103	0.8363	0.9101	0.8362	0.9108	0.8331	0.9861	0.9751
KNN	0.9109	0.8403	0.9091	0.8409	0.9127	0.8414	0.9089	0.8353	0.9704	0.9492
XGBoost	0.8882	0.8057	0.8872	0.8062	0.8871	0.8068	0.8882	0.8021	0.9812	0.9687
Decision Tree	0.8711	0.7976	0.8705	0.7960	0.8703	0.7945	0.8711	0.7945	0.9141	0.8804
SVM	0.8539	0.7627	0.8513	0.7647	0.8522	0.7668	0.8539	0.7582	0.9703	0.9627
AdaBoost	0.7291	0.5758	0.7275	0.5835	0.7325	0.5901	0.7291	0.5728	0.9068	0.8728
Logistic Regression	0.6321	0.5313	0.6244	0.5270	0.6241	0.5229	0.6321	0.5258	0.8409	0.8542
Naive Bayes	0.5329	0.4361	0.5099	0.4342	0.5194	0.4314	0.5329	0.4326	0.7721	0.7998

TABLE 3. Performance of classification algorithms on M4 and M7 datasets.

Model	Hamming loss		Jaccard Score		Subset Accuracy		F1 Micro		F1 Macro	
	M4	M7	M4	M7	M4	M7	M4	M7	M4	M7
Extra Trees	0.0499	0.1229	0.9062	0.7806	0.9501	0.8770	0.9500	0.8770	0.9497	0.8751
LightGBM	0.0579	0.1167	0.8925	0.7905	0.9420	0.8832	0.9420	0.8832	0.9419	0.8810
Random Forest	0.0623	0.1269	0.8841	0.7741	0.9376	0.8731	0.9376	0.8731	0.9372	0.8706
Neural Network	0.0675	0.1431	0.8757	0.7481	0.9324	0.8569	0.9324	0.8569	0.9318	0.8530
Gradient Boosting	0.0892	0.1636	0.8389	0.7181	0.9107	0.8363	0.9107	0.8363	0.9102	0.8327
KNN	0.0890	0.1596	0.8372	0.7223	0.9109	0.8403	0.9109	0.8403	0.9089	0.8357
XGBoost	0.1117	0.1942	0.8019	0.6734	0.8882	0.8057	0.8882	0.8057	0.8871	0.8011
Decision Tree	0.1353	0.2023	0.7666	0.6652	0.8646	0.7976	0.8646	0.7976	0.8640	0.7943
SVM	0.1461	0.2372	0.7480	0.6149	0.8538	0.7627	0.8538	0.7627	0.8512	0.7569
AdaBoost	0.2709	0.4242	0.5803	0.4067	0.7290	0.5758	0.7290	0.5758	0.7274	0.5747
Logistic Regression	0.3678	0.4686	0.4633	0.3579	0.6321	0.5313	0.6321	0.5313	0.6244	0.5203
Naive Bayes	0.4671	0.5639	0.3529	0.2714	0.5328	0.4361	0.5328	0.4361	0.5099	0.4232

TABLE 4. Mean and standard deviation ($\mu \pm \sigma$) of performance indicators on M4 and M7 dataset.

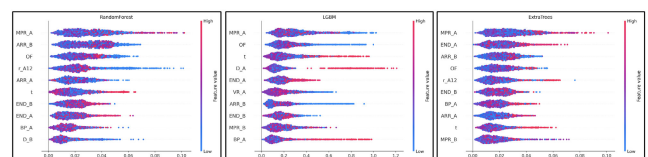
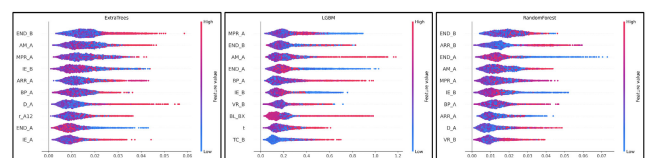
Model	Accuracy		F1 Score		ROC-AUC	
	M4	M7	M4	M7	M4	M7
Extra Trees	0.9501 $\pm$ 0.0055	0.8771 $\pm$ 0.0015	0.9498 $\pm$ 0.0056	0.8765 $\pm$ 0.0016	0.9953 $\pm$ 0.0007	0.9779 $\pm$ 0.0008
LightGBM	0.9421 $\pm$ 0.0047	0.8832 $\pm$ 0.0019	0.9419 $\pm$ 0.0048	0.8830 $\pm$ 0.0019	0.9932 $\pm$ 0.0010	0.9859 $\pm$ 0.0002
Random Forest	0.9376 $\pm$ 0.0054	0.8731 $\pm$ 0.0016	0.9373 $\pm$ 0.0050	0.8728 $\pm$ 0.0016	0.9931 $\pm$ 0.0008	0.9827 $\pm$ 0.0005

TABLE 5. Multilabel classification accuracies.

Crystal System	Actual count	Random Forest	Extra Trees	Light GBM
Cubic	745	730	731	729
Hexagonal	20	19	20	19
Monoclinic	28	27	28	27
Orthorhombic	350	337	338	339
Triclinic	14	13	14	13
Trigonal	92	87	90	86
Tetragonal	33	32	33	31
Overall Accuracy		97.11%	97.81%	97.03%

of each feature to the prediction. It provides both local interpretability by explaining individual predictions and global interpretability by understanding the overall feature effects.

SHAP analysis was done on the best three classifiers, namely Extra Trees, LightGBM and Random Forest, for M4 and M7 data sets. The 10 most prominent features were identified for each classifier. Among these features, the average SHAP values for each feature from the three classifiers were taken for M4 and M7 data sets. A description

**FIGURE 8.** Feature contributions and distribution for M4 dataset.**FIGURE 9.** Feature contributions and distribution for M7 dataset.

of the SHAP identified features for the Averaged SHAP values, their feature contributions and distribution across the samples for both data sets are given in Figures 8, 9, 10 and 11.

Across both M4 and M7 datasets, the most important features consistently reflect the geometric and electronic

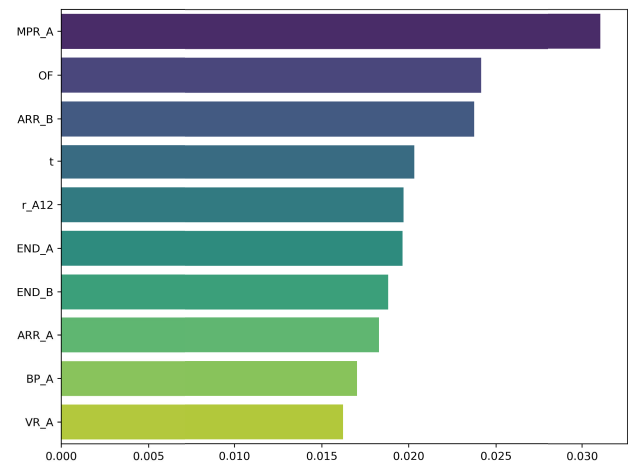


FIGURE 10. Averaged feature importance in M4 dataset.

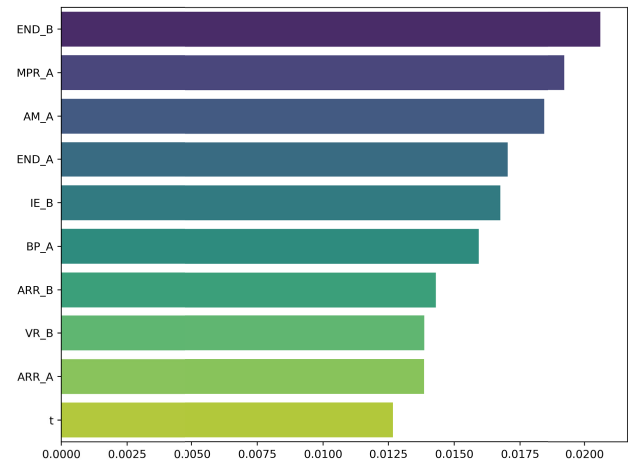


FIGURE 11. Averaged feature importance in M7 dataset.

factors governing perovskite crystal symmetry. Size-related descriptors, including atomic radius ratios and the tolerance factor (t), highlight the role of ionic size mismatch and octahedral tilting in driving symmetry lowering in accordance with Goldschmidt-type stability rules. Electronic features, particularly electronegativity differences at the B site (END_B), capture variations in B–O bond character that influence octahedral distortions and lattice energetics. Additional A-site thermophysical properties, such as melting point and mass-related descriptors, modulate structural rigidity and stability across different crystal systems. Together, these features encode the coupled structural and electronic constraints that determine perovskite crystal system stability.

$M4^*$ and $M7^*$ are the newer versions of the $M4$ and $M7$ datasets that contain only the 10 prominent features obtained from averaged SHAP analysis. A fivefold cross-validated data classification was performed with the Random Forest (RF), Extra Trees (ET) and LightGBM (LGBM) classifiers on both the $M4^*$ and $M7^*$ data sets. It was observed that the accuracy results were comparable with the results obtained using all the features, and Random Forest and LightGBM

TABLE 6. Performance evaluation of algorithms on $M7^*$, $M7$, $M4^*$ and $M4$ data sets.

Model	M7*		M7		M4*		M4	
	Acc	Time	Acc	Time	Acc	Time	Acc	Time
ET	87.65	8.61	87.71	58.85	94.04	4.07	95.01	4.73
LGBM	87.39	6.17	88.32	10.17	92.35	2.37	94.21	5.01
RF	87.25	36.62	87.31	11.19	93.74	17.51	93.76	43.87

classifiers took considerably less time. A comparison of the accuracy in percentage and the time taken in seconds is given in Table 6.

Based on these results, the performance of the reduced-feature datasets ($M4^*$ and $M7^*$) can be interpreted as an accuracy–efficiency trade-off that is highly favourable for high-throughput screening applications. In particular, the reduced $M4^*$ dataset shows only a minor drop in accuracy of approximately 1.5–2% across the classifiers when compared to the full $M4$ dataset, while simultaneously achieving a substantial reduction in inference time, exceeding 50% for Random Forest and LightGBM. Such reductions in computational cost become critical when screening millions of hypothetical compositions, where marginal losses in accuracy are outweighed by orders-of-magnitude gains in throughput. Consequently, the reduced-feature models offer a highly practical and scalable solution for real-world materials discovery pipelines, enabling rapid exploration of large compositional spaces without significant compromise in predictive reliability.

VI. CONCLUSION AND FUTURE WORK

We have conducted a multi-class, multi-label data classification to predict the crystal system of perovskites using elementary properties of the elements in A and B sites. Extra Trees, LightGBM and Random Forest algorithms performed well above the other algorithms for both multiclass and multilabelled classification. These algorithms predicted the crystal systems with an average accuracy of 88% and 94% on $M7$ and $M4$ datasets during multiclass classification. For the multilabelled classification, these algorithms could predict the possible crystal systems with an accuracy of above 95% consistently. SHAP analysis indicates the dominance of the features generated using feature engineering. The averaged top ten features of the top performing classification algorithms could capture the physics underlying the crystal system of perovskite materials with almost the same accuracies in considerably less time. The strong performance of tree-based models reflects their ability to effectively capture the nonlinear relationships, threshold effects, and coordination-driven constraints inherent in perovskite chemistry. Their interpretability through SHAP enables the extraction of physically meaningful trends, showing that learnt feature importance corresponds well with established crystallographic knowledge.

Predicting crystal systems can be used as a starting point in understanding the structure of perovskites. Advanced and optimised machine learning algorithms, along with enhanced feature engineering techniques and high-quality datasets,

can outperform computational methods, including density functional theory (DFT), which is both time-consuming and computationally expensive. Machine learning, through its diverse algorithmic and software ecosystems, represents a paradigm shift in predicting perovskite crystal systems based on elemental properties. This data-driven approach not only offers rapid, cost-effective, and accurate predictions but also fosters a deeper understanding of the underlying chemical and physical principles governing crystal structure formation, thereby significantly contributing to Accelerated Materials Discovery, Unveiling Structure-Property Relationships, Reduced Experimental Costs and Guided Synthesis, Leveraging Large Materials Databases, Prediction of Metastable Phases, and Rational Material Design.

The present work is limited by its reliance on composition-only features, which cannot fully represent temperature-, pressure-, or defect-induced structural transitions. Furthermore, dataset biases originating from DFT-derived entries in the Materials Project restrict the model's applicability to experimentally stabilised or metastable phases that are under-represented in computational databases. Future extensions of this work include incorporating experimental data to reduce simulation bias, expanding predictions to spacegroup-level symmetry, and modelling environmental dependencies such as temperature, pressure, and strain that influence phase stability. Investigating Graph Neural Networks (GNNs), which provide structural information, may improve predictive capabilities and facilitate hybrid composition-structure models. Additionally, integrating the trained classifiers into inverse design workflows could accelerate the targeted discovery of perovskite materials for optoelectronics, photovoltaics, and other energy applications.

APPENDIX A SUPPLEMENTARY INFORMATION

The data sets and python code used in this paper are available on github at https://github.com/medackan/CSP_Perovskites.
git.

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